Lab Assignment 08

BRAC UNIVERSITY

Inspiring Excellence

Course Code:	**CSE111**
Course Title:	**Programming Language II**
Topic:	**Static Variable & Static Method**
Number of Tasks:	**11**

[You are not allowed to change the driver codes of any of the tasks]

Task 1

Design the **Passenger** class in such a way that the following code provides the expected output.

- Passenger class has two static variables ***no_of_passenger*** and ***total_fare.***
- Each passenger has to pay 20 TK/Distance and extra 10 TK/BaggageWeight.

Given Code	Expected Output

`public class PassengerTester{`
`  public static void main(String args[]){`
`  System.out.println("Total Passenger: "+ Passenger.no_of_passenger);`
`  System.out.println("Total Fare: "+ Passenger.total_fare + " TK");`
`  System.out.println("==========1==========");`
`  Passenger p1 = new Passenger("Lara", 5.6);`
`  p1.passengerDetails();`
`  System.out.println("==========2==========");`
`  Passenger p2 = new Passenger("Kevin", 10.0);`
`  p2.setBaggageWeight(6.8);`
`  p2.passengerDetails();`
`  System.out.println("==========3==========");`
`  Passenger p3 = new Passenger("Robin", 2.3);`
`  p3.setBaggageWeight(5);`
`  p3.passengerDetails();`
`  System.out.println("==========4==========");`
`  System.out.println("Total Passenger: "+ Passenger.no_of_passenger);`
`  System.out.println("Total Fare: "+ Passenger.total_fare + " TK");`
`  }`
`}`
	`Total Passenger: 0`
	`Total Fare: 0.0 TK`
	`==========1==========`
	`Name: Lara`
	`Fare: 112.0 TK`
	`==========2==========`
	`Name: Kevin`
	`Fare: 268.0 TK`
	`==========3==========`
	`Name: Robin`
	`Fare: 96.0 TK`
	`==========4==========`
	`Total Passenger: 3`
	`Total Fare: 476.0 TK`

Task 2

Design a **Book** class in such a way that the following code provides the expected output.

- The Book class has two static variables: total_books_sold and total_revenue.
- Each book has a base price of 150 TK. If the discountPercentage is applied, the book's price is reduced by that percentage.
- The Book class should have a method to calculate the price after the discount

Given Code

```
public class BookTester {
    public static void main(String[] args) {
        System.out.println("Total Books Sold: " + Book.total_books_sold);
        System.out.println("Total Revenue: "+Book.total_revenue + " TK");
        System.out.println("==========1==========");

        Book b1 = new Book("Java Programming", 10); // 10% discount
        b1.bookDetails();

        System.out.println("==========2==========");

        Book b2 = new Book("Python Programming", 15); // 15% discount
        b2.bookDetails();

        System.out.println("==========3==========");

        Book b3 = new Book("Data Structures", 5); // 5% discount
        b3.bookDetails();

        System.out.println("==========4==========");
        System.out.println("Total Books Sold: " + Book.total_books_sold);
        System.out.println("Total Revenue: "+Book.total_revenue + " TK");
    }
}
```

Expected Output

```
Total Books Sold: 0
Total Revenue: 0.0 TK
==========1==========
Title: Java Programming
Price after Discount:
135.0 TK
==========2==========
Title: Python Programming
Price after Discount:
127.5 TK
==========3==========
Title: Data Structures
Price after Discount:
142.5 TK
==========4==========
Total Books Sold: 3
Total Revenue: 405.0 TK
```

Task 3

Design a **Student** class in such a way that the following code provides the expected output.

Driver Code	Output

`public class StudentTester {`
`  public static void main(String[] args) {`
`  Student.printDetails();`
`  System.out.println("----------");`
`  Student mikasa = new Student("Mikasa", 3.75);`
`  mikasa.individualDetail();`
`  System.out.println("----------");`
`  Student.printDetails();`
`  System.out.println("----------");`
`  Student harry = Student.createStudent("Harry", 2.5, "Charms");`
`  harry.individualDetail();`
`  System.out.println("----------");`
`  Student.printDetails();`
`  System.out.println("----------");`
`  Student levi = new Student("Levi", 3.33);`
`  levi.individualDetail();`
`  System.out.println("----------");`
`  Student.printDetails();`
`  }`
`}`
	`Total Student(s): 0`
	`CSE Student(s): 0`
	`Other Department Student(s): 0`
	`----------`
	`ID: 1`
	`Name: Mikasa`
	`CGPA: 3.75`
	`Department: CSE`
	`----------`
	`Total Student(s): 1`
	`CSE Student(s): 1`
	`Other Department Student(s): 0`
	`----------`
	`ID: 2`
	`Name: Harry`
	`CGPA: 2.5`
	`Department: Charms`
	`----------`
	`Total Student(s): 2`
	`CSE Student(s): 1`
	`Other Department Student(s): 1`
	`----------`
	`ID: 3`
	`Name: Levi`
	`CGPA: 3.33`
	`Department: CSE`
	`----------`
	`Total Student(s): 3`
	`CSE Student(s): 2`
	`Other Department Student(s): 1`

Task 4

Suppose you have opened a new library, from where your friends can borrow books. Initially you have bought 3 books (Pather Panchali, Durgesh Nandini & Anandmath) each of 3 copies only. Design the **Borrower** class in such a way that the following code provides the expected output.

- You are given the arrays ***book_count*** and ***book_name*** to keep track of the number of books available. For simplicity, assume that there will be no other books in the library.
- You must reuse the ***remainingBooks()*** method when needed.

Given Code

```
public class Tester{
  public static void main(String args[]){
    Borrower.bookStatus();
    System.out.println("*********1*********");
    Borrower b1 = new Borrower("Nabila");
    b1.borrowBook("Pather Panchali");
    b1.borrowBook("Anandmath");
    b1.borrowerDetails();
    System.out.println("*********2*********");
    Borrower b2 = new Borrower("Sadia");
    b2.borrowBook("Anandmath");
    b2.borrowBook("Durgesh Nandini");
    b2.borrowBook("Pather Panchali");
    b2.borrowerDetails();
    System.out.println("*********3*********");

System.out.println(Borrower.remainingBooks("Anandmath")+"
copies of Anandmath is remaining.");
    System.out.println("*********4*********");
    Borrower b3 = new Borrower("Anika");
    b3.borrowBook("Anandmath");
    Borrower.bookStatus();
    System.out.println("*********5*********");
    Borrower b4 = new Borrower("Oishi");
    b4.borrowBook("Anandmath");
    b4.borrowBook("Durgesh Nandini");
    b4.borrowerDetails();
  }
}

public class Borrower{
  public static int book_count[] = {3, 3, 3};
  public static String book_name[] = {"Pather Panchali",
"Durgesh Nandini", "Anandmath"};

  // Your Code here
}
```

Expected Output

```
Available Books:
Pather Panchali: 3
Durgesh Nandini: 3
Anandmath: 3
*********1*********
Name: Nabila
Books Borrowed:
Pather Panchali
Anandmath
*********2*********
Name: Sadia
Books Borrowed:
Anandmath
Durgesh Nandini
Pather Panchali
*********3*********
1 copies of Anandmath is
remaining.
*********4*********
Available Books:
Pather Panchali: 1
Durgesh Nandini: 2
Anandmath: 0
*********5*********
This book is not available.
Name: Oishi
Books Borrowed:
Durgesh Nandini
```

Task 5

For this task, you need to design the **Cargo** class with appropriate static and non-static variables and methods to produce this given output for the given tester code.

Note: .load() method marks an object as selected for transport, and .unload() method unmarked it. At a time, the transport capacity is 10.0 Tonnes. Each Cargo object is initialized with 2 attributes from the constructor - the contents and the weight. Carefully observe the outputs to identify the other attributes and design the class.

Given Code

```
public class CargoTester {
  public static void main(String[] args) {
    System.out.println("Cargo Capacity: " + Cargo.capacity());
    System.out.println("1====================");
    Cargo a = new Cargo("Industrial Machinery", 4.5);
    a.details();
    System.out.println("2====================");
    a.load();
    System.out.println("3====================");
    Cargo b = new Cargo("Steel Ingot", 2.7);
    b.details();
    System.out.println("4====================");
    System.out.println("Cargo Capacity: " + Cargo.capacity());
    System.out.println("5====================");
    b.load();
    System.out.println("Cargo Capacity: " + Cargo.capacity());
    System.out.println("6====================");
    Cargo c = new Cargo("Tree Trunks", 3.6);
    c.load();
    System.out.println("7====================");
    c.details();
    b.details();
    System.out.println("8====================");
    Cargo d = new Cargo("Processed Goods", 1.8);
    d.load();
    System.out.println("Cargo Capacity: " + Cargo.capacity());
    System.out.println("9====================");
    b.unload();
    System.out.println("Cargo Capacity: " + Cargo.capacity());
    System.out.println("10====================");
    c.load();
    System.out.println("11====================");
    b.details();
    System.out.println("Cargo Capacity: " + Cargo.capacity());
  }
}
```

Expected Output

```
Cargo Capacity: 10.0
1====================
Cargo ID: 1, Contents: Industrial
Machinery, Weight: 4.5, Loaded: false
2====================
Cargo 1 loaded for transport.
3====================
Cargo ID: 2, Contents: Steel Ingot,
Weight: 2.7, Loaded: false
4====================
Cargo Capacity: 5.5
5====================
Cargo 2 loaded for transport.
Cargo Capacity: 2.8
6====================
Cannot load cargo, exceeds weight
capacity.
7====================
Cargo ID: 3, Contents: Tree Trunks,
Weight: 3.6, Loaded: false
Cargo ID: 2, Contents: Steel Ingot,
Weight: 2.7, Loaded: true
8====================
Cargo 4 loaded for transport.
Cargo Capacity: 1.0
9====================
Cargo 2 unloaded.
Cargo Capacity: 3.7
10====================
Cargo 3 loaded for transport.
11====================
Cargo ID: 2, Contents: Steel Ingot,
Weight: 2.7, Loaded: false
Cargo Capacity: 0.09999999999999964
```

Task 6

Complete the class **Circle** so that the desired outputs are generated properly.

Given Code	Expected Output

`public class shapeTester {`
`  public static void main(String[] args) {`
`  Circle c = new Circle();`
`  System.out.println("=====================");`
`  c.name = "Circle";`
`  c.color = "Red";`
`  c.radius = 5;`
`  c.displayInfo();`
`  System.out.println("=====================");`
`  c.area();`
`  }`
`}`
 	
`public class Shape {`
`  public String name;`
`  public String color;`
 	
`  public void displayInfo() {`
`  System.out.printf("Name: %s\nColor: %s\n", name, color);`
`  }`
`}`
 	
`public class Circle extends Shape {`
`  //Your Code Here`
`}`
	`=====================`
	`Name: Circle`
	`Color: Red`
	`=====================`
	`Area of Red Circle: 78.54`

Task 7

Complete the class **Dog** so that the desired outputs are generated properly.

Given Code	Expected Output

`public class AnimalTester{`
`  public static void main(String args[]){`
`  Animal a1 = new Animal();`
`  System.out.println("1-------------");`
`  a1.details();`
`  System.out.println("2-------------");`
`  Dog d1  = new Dog();`
`  d1.name = "Pammy";`
`  System.out.println("3-------------");`
`  System.out.println("Name: " + d1.getName());`
`  d1.details();`
`  System.out.println("4-------------");`
`  d1.updateSound("Bark");`
`  System.out.println("5-------------");`
`  d1.details();`
`  }`
`}`
 	
`public class Animal{`
`  public int legs = 4;`
`  public String sound = "Not defined";`
 	
`  public void details(){`
`  System.out.println("Legs: "+legs);`
`  System.out.println("Sound: "+sound);`
`  }`
`}`
 	
`public class Dog extends Animal{`
`  //Your Code Here`
`}`
	`1-------------`
	`Legs: 4`
	`Sound: Not defined`
	`2-------------`
	`The dog says hello!`
	`3-------------`
	`Name: Pammy`
	`Legs: 4`
	`Sound: Not defined`
	`4-------------`
	`5-------------`
	`Legs: 4`
	`Sound: Bark`

Task 8

		Output	
1.	public class Maze{		
2.	public static int x;		
3.	public void methodA(){		
4.	int m = 5;		
5.	x=11;		
6.	System.out.println(x+" "+m);		
7.	m=methodB(m-3)+x;		
8.	System.out.println(x+" "+(m));		
9.	methodB(x,m);		
10.	System.out.println(x+" "+m+x);		
11.	}		
12.	public int methodB(int y){		
13.	x=y*y;		
14.	System.out.println(x+" "+y);		
15.	return x+3;		
16.	}		
17.	public void methodB(int z, int x){		
18.	z=z-2;		
19.	x=x*1%z;		
20.	System.out.println(z+" "+x);		
21.	}		
22.	}		
23.	public class Test8{		
24.	public static void main(String [] args){		
25.	Maze c = new Maze();		
26.	c.methodA();		
27.	c.methodB(-11, 45);		
28.	}		
29.	}		

Task 9

		Output		
1.	**public class Tracing {**			
2.	public static int x= 0, y = 0;			
3.	public int a, b;			
4.	public Tracing(int a, int b){			
5.	this.a = a;			
6.	this.b = b;			
7.	x+=1;			
8.	y+=2;			
9.	}			
10.	public void methodA(int a){			
11.	this.a = x+a;			
12.	this.b = this.b+ this.a +this.methodB();			
13.	System.out.println(this.a+" "+this.b+" "+x);			
14.	}			
15.	public int methodB(){			
16.	this.b = y - this.b + this.a;			
17.	System.out.println(this.a+" "+this.b+" "+x);			
18.	x += this.b;			
19.	return this.b;			
20.	}			
21.	public void methodB(Tracing t1){			
22.	t1.b = this.y - t1.b + this.b;			
23.	System.out.println(t1.a+" "+t1.b+" "+x);			
24.	}			
25.	}			
26.	**public class Test9{**			
27.	public static void main(String [] args){			
28.	Tracing t1= new Tracing(2, 3);			
29.	t1.methodA(1);			
30.	Tracing t2= new Tracing(3, 4);			
31.	t2.methodA(2);			
32.	t1.methodB(t2);			
33.	t2.methodB(t2);			
34.	}			
35.	}			

Task 10

		Outputs		
1	**public class FinalT6A{**			
2	public static int temp = 3;			
3	public int sum;			
4	public int y = 2;			
5	public FinalT6A(int x, int p){			
6	temp+=3;			
7	y = temp - p;			
8	sum = temp + x;			
9	System.out.println(x + " " + y+ " " + sum);			
10	}			
11	public void methodA(){			
12	int x=0, y =0;			
13	y = y + this.y;			
14	x = this.y + 2 + temp;			
15	sum = x + y + methodB(temp, y);			
16	System.out.println(x + " " + y+ " " + sum);			
17	}			
18	public int methodB(int temp, int n){			
19	int x = 0;			
20	y = y + (++temp);			
21	x = x + 2 + n;			
22	sum = sum + x + y;			
23	System.out.println(x + " " + y+ " " + sum);			
24	return sum;			
25	}			
26	}			
27	**public class Test10{**			
28	public static void main(String [] args){			
29	FinalT6A q1 = new FinalT6A(2,1);			
30	q1.methodA();			
31	q1.methodA();			
32	}			
33	}			

Task 11

Find the outputs after running the main() method in **Test11** class.

		Outputs		
1	**public class Quiz1{**			
2	public static int temp = 4;			
3	public int sum;			
4	public int y;			
5	public Quiz1(){			
6	y = temp - 1;			
7	sum = temp + 1;			
8	temp+=2;			
9	}			
10	public Quiz1(int p){			
11	y = temp + p ;			
12	sum = p + temp + 1;			
13	temp-=1;			
14	}			
15	public void methodA(){			
16	int x=0, y =0;			
17	y = y + this.y;			
18	x = this.y + 2 + temp;			
19	sum = x + y + methodB(x, y);			
20	System.out.println(x + " " + y+ " " + sum);			
21	}			
22	public int methodB(int m, int n){			
23	int x = 0;			
24	y = y + m + (++temp);			
25	x = x + 2 + n;			
26	sum = sum + x + y;			
27	System.out.println(x + " " + y+ " " + sum);			
28	return sum;			
29	}			
30	}			
31	**public class Test11{**			
32	public static void main(String [] args){			
33	Quiz1 q1 = new Quiz1();			
34	q1.methodA();			
35	q1.methodA();			
36	Quiz1.temp+= 2;			

37	`        Quiz1 q2 = new Quiz1(2);`	
38	`        q2.methodA();`	
39	`        q2.methodA();`	
40	`      }`	
41	`}`	